

Stuyvesant Faculty Perspectives:
Mr. JonAlf Dyrland-Weaver, Class of 2001:
Computer Science Principles, in Class and in Daily Life

Craig Chen



Mr. JonAlf Dyrland-Weaver has inspired students in computer science for 18 years. After graduating from Stuyvesant in 2001, he majored in Philosophy and minored in Computer Science at Columbia College. He then earned a master's degree in Mathematics Education from the City College of New York. In 2005, he began his career as a teacher at Brandeis High School through the New York City Teaching Fellows Program. At Stuyvesant, he has taught a wide range of courses including Intro to Computer Science, AP Computer Science, Software Development, Systems Level Programming, and Computer Graphics, and has expanded Stuyvesant's computer science department to include courses such as NeXTCS. Mr. DW, as he is popularly known, currently serves as Stuy's Computer Science Coordinator.

Mr. Dyrland-Weaver's answers are taken from an interview conducted by Craig Chen and have been edited for clarity.

1) Why do you enjoy computer science and what makes teaching it meaningful to you?

Computer science is, to me, about solving problems and looking at problems in interesting ways. How do I get information that is useful to people in an accessible and understandable way? Or how do I make a fun game? These are practical problems, but there are also theoretical problems that have to do with questions of computability and complexity. Computer science runs the gamut of problem solving, and in my own eyes, that's what makes the field so rich.

When I teach computer science, I want to portray the learning of the subject as building up a toolbox of techniques, programs, and environments, as well as the understanding of when to

use them. I hope to empower others to develop and use such tools to do other things—to do interesting things. At the end of the day, I am giving my students the ability to solve all sorts of different kinds of problems with computers, which are incredibly useful and powerful tools. The better that my students understand how to use them, the better they're going to be able to do whatever it is, whether it's write software, produce webpages, or do anything else. And so that's what it is for me, and that's why I enjoy teaching. It provides students with freedom of exploration.

2) What do you believe is one of the most underappreciated aspects of being a computer science teacher?

It would be the amount of time and thought that goes into creating good problems, assignments, and labs. Creating challenging and rich study material in computer science is different from any other subject, let's say, math. And it's not to say that math teachers don't spend a lot of time crafting problems as well. I've seen Mr. Kats and Mr. Cocoros sit down and really work things out for a whole period to create a good problem. I'm choosing it because I taught it for a while. Say you learn about the Pythagorean Theorem. It is helpful to practice the theorem a bunch of times. However, in computer science, if I want students to write a function to find the hypotenuse of a right triangle, it isn't really meaningful for me to have them repeat the task. It would just result in them writing the same code. But writing the same thing multiple times in CS has a high diminishing marginal return, and I generally have a rule that I never have my students write the same thing more than once. Therefore, coming up with problems in computer science is difficult because it requires creating bigger frameworks so that students have both room to explore and practice.

This concept has come up a lot for me recently with the NeXTCS class, which is a new course. I had my students create sorting algorithms on Processing that visually moved blocks around based on their sizes. The purpose of the assignment was for students to learn and write the sorting algorithms, and I had to prepare all of the visual stuff to ensure it would work. As long as the students wrote their sorting algorithm correctly, they could focus on the thing that they were supposed to be learning. Though that's just a particular example; this shows up in CS classes all over. Mr. Holmes is famous for spending lots and lots of time producing homework because he wants it to be the right thing for students to work on. The amount of work that goes into producing good assignments is great, even if they're small assignments. So maybe that is the most unappreciated piece.

3) Could you ever see yourself teaching another subject or working in a completely different job?

Teaching a different subject? No. My teaching career started 17 and a half years ago as a math teacher because that was the only avenue available to me. Then, 15 and a half years ago, I came to Stuy, switched to teaching computer science full time, and never looked back.

There are plenty of subjects that I would enjoy learning. I was a philosophy major in college, so I'm not a stranger to the humanities. But in terms of teaching, I love teaching computer science. I couldn't see myself teaching anything else. Even if I teach the same classes ten years in a row, such as graphics, there are always new programming challenges that keep the teaching interesting and enjoyable. It's not like I'm just opening up my old lessons and teaching the same things.

In terms of seeing myself do something else for a living, probably not. I do enjoy programming, so maybe there's a world where I could do some software development. Nonetheless, despite all of the problems that exist in the teaching profession, I enjoy it far too much to leave it in this lifetime. I'm not just writing code; I'm interacting with students. If you're a teacher and you're doing it for long enough, you're sticking around usually because there's something about being in a room with 30 students and working with them that is rewarding. It's not the vacations, it's not any of the silly things. It's working with students.

4) What's one advice you would give to students? It doesn't have to be specific to computer science.

Well, I think a piece of computer science advice that is also a good piece of life advice is that there is very rarely a "best" anything. Students will ask, "Oh, what's the best programming language? What's the best kind of computer? What's the best code editor?" The answer is that nothing is the best. Certainly, some are better for specific tasks, but there are far too many of any of those things for any one of them to be supremely the best. There was a period of time when I was younger when I would fall into this trap of being like, "Oh, well, this is clearly a better one." But in reality, you should recognize the good qualities in any of these tools, or whatever they might be. And also recognize that just because someone prefers one over the other does not define their own worth. Just because someone really likes programming in Java doesn't make them any better or worse than someone who really likes to program in Python. Saying this in the context of computer science is important, but it extends to all of life in general. Rather than focusing on what is best, we should concentrate on learning about a wide range of experiences and understanding.

The *Star Wars* phrase "Only a Sith deals in absolutes" comes to mind. The world is complex. There are good and bad things about all sorts of things. We should be able to recognize that and find it worth having, regardless of your own personal feelings. No matter their preferred subject—musical tastes, movie tastes, or any of that—everyone has value we should appreciate.

5) Through your own experiences teaching at Stuy for over 15 and half years, how have you seen the school change?

The 2007-2008 school year was my first year teaching here, and shortly before that time I was a student here, graduating in 2001. So I've been here for a total of nearly 20 years. One of the positive ways that Stuy has changed is that students have become more accepting of different experiences and different types of people. I think that there is more of a sense of people being able to find their place. Stuyvesant has always had more than 80,000 clubs and 40 sports teams, providing a whole range of different communities where students can find their interest and expand it. Stuy Squad is something that comes to mind; that didn't exist when I started teaching here. It's this thing everybody enjoys doing, and I don't know the real evolution of it, but I imagine people really enjoyed doing the dance stuff in SING! and made it a whole separate event.

However, during that same period, the level of competitiveness and focus on grades increased to the detriment of learning. Though it has always existed, it seems harder for students not to get sucked into the competitive atmosphere. Everybody thinks that they've got to be better than the other students in order to get ahead in class, in college, in life and so on. I think that that has gotten somewhat worse over time.

6) How do you think your own teaching has changed over 17 years?

I've improved my ability to listen to students and determine the value in the way they see problems or potential solutions, even if they aren't the things I expect to be included. For example, let's say I have a lesson about binary search. I first want to have a discussion about it, and eventually get to talking about its actual algorithm. Earlier in my career, when students would have that type of open discussions, I was more likely to brush those off instead of giving their ideas a moment to be in the class and be heard. I've gotten better at respecting ideas that are not what I would have expected or where I wanted lessons to go.

I've really been trying to promote a judgment-free zone, more so as my career has progressed. I happen to very much enjoy *Star Wars* and *Star Trek*, and early on in my teaching career, if the student said, "Oh, I've never seen *Star Wars*," I would be like, "What do you mean you've never seen *Star Wars*?" The fact of the matter is, it doesn't matter whether or not you've seen *Star Wars*. It doesn't affect whether you'd make a good computer science student or teacher. Now when I teach, I make it clear that it is something that I enjoy, but it doesn't have to be this "I can't believe you've never seen this." I don't want those sorts of unofficial barriers, and I work to ensure I am not creating any myself.

7) How do you spend your time outside of teaching? What do you enjoy doing the most?

Well, what I spend the most time doing right now is spending time with my son. I have a three-year-old, and that is all the time that I have, for the most part, in terms of my free time. And it's wonderful, but it is hard and difficult. Readers of this publication are young, so you're probably not looking into having children anytime soon, but don't do it quickly. It's a lot of work, even though it is rewarding.

But if I'm not hanging out with my son, I'd like to spend my time with friends playing board games, like tabletop role-playing games. I have a group that I've been playing pen and paper role-playing games with for almost three years now. We played a *Star Wars* game for a couple of years, and now we're doing a superhero-style game, such as *Dungeons and Dragons*. And in a similar vein, there's a phenomenal *Star Wars* card game that I had a really good group of people to play with. There was a time when I was competitive nationally. It's not a massive game like *Magic*, *The Gathering*, or *Pokemon*, but it's something I spent a lot of time playing and became quite skilled at.